HERIOT WATT UNIVERSITY

SCHOOL OF MATHEMATICAL AND COMPUTER SCIENCES

Department of Actuarial Mathematics & Statistics

F78AB

ACTUARIAL & FINANCIAL MATHEMATICS B

Semester 2 2018/19

Duration: Two Hours

Total marks: 60

ANSWER ALL SIX QUESTIONS

UNIVERSITY APPROVED CALCULATOR MAY BE USED

Q1 An investor is considering purchasing a solar power plant. Over the course of a year the plant is expected to generate 120,000MWh of electricity, which the investor assumes is generated at a constant rate over the year. The expected life of the plant is 20 years. The investor expects to receive two payments for the electricity:

- Payments from sale to the national grid of £20 per MWh payable continuously.
- A government subsidy payable at the end of each year for the electricity generated that year. This is set at £40 per MWh for 2020 and is index-linked, with the increase each year in line with the change in RPI for the year to the November preceding the year end.

The investor wishes to make the purchase on 1 January 2020, so they receive the full payments for 2020. The investor assumes inflation will increase at 2.5% per annum and they will know the November 2019 RPI at the time of purchase.

(a) Assuming the investor requires a minimum return of 7% per annum effective, calculate the maximum price the investor will pay for the plant.

(5 marks)

The investor manages to buy the plant for £82 million and expects to sell it exactly five years later for £72 million.

(b) Calculate the expected real return on this investment.

(6 marks)

(Total = 11 marks)

Q2 The rate of return on a stock in each future year is assumed to be independent and identically distributed with mean 6% and variance 0.02. Deriving all necessary formulae, calculate each of the following:

(a) The expected accumulation after ten years of one unit invested in the stock.

(3 marks)

(b) The variance of the accumulation after ten years of one unit invested in the stock.

(5 marks)

(c) What difference would it make the answers in (a) and (b) if the initial investment was in ten units of the stock.

(1 mark)

(Total 9 marks)

Q3 The force of interest, $\delta(t)$, is a function of time and, at any time *t*, is given by the formula:

$$\delta(t) = \begin{cases} 0.05 & 0 \leq t \leq 3 \\ 0.08 - 0.01t & 3 < t \leq 6 \\ 0.01t - 0.04 & 6 < t \end{cases}$$

(a) Calculate the value at time $t = 4$ of £1,000 due for payment at time t = 8.

(5 marks)

(b) Calculate the constant rate of interest rate per annum convertible monthly which leads to the same result as in (a) being obtained.

(2 marks)

(c) Calculate the accumulated amount at time $t = 10$ of a payment stream, paid continuously from time $t = 0$ to time $t = 3$ under which the payment rate at time *t* is $\rho(t) = 100e^{0.03t}$.

(6 marks)

(Total 13 marks)

PLEASE TURN OVER

Q4 A company has a liability of £200,000 due in 10 years time. The company has just exactly the funds to pay the liability, using the current market effective interest rate of 7% per annum. This is also the interest rate earned on cash.
Because of regulatory requirements, the company is required to hold 10% of its funds in cash. It can invest the balance in the following securities:

- A zero-coupon bond redeemable at par in 12 years time.
- A fixed-interest bond which is redeemable at 105% in 16 years time, with a coupon of 5% per annum payable annually in arrear.

(a) Calculate the nominal amounts of the bonds which need to be purchased to satisfy Redington's first two conditions for immunisation.

(10 marks)

(b) Calculate the amount which should be invested in each of the bonds.

(2 marks)

(c) Without doing any further calculations, explain whether the company would be immunised against small changes in the rate of interest if the quantities of each bond calculated in part (a) are purchased.

(2 marks)

(Total 14 marks)

Q5

(a) Define a replicating portfolio for a contract.

(1 mark)

(b) A share will pay a dividend of 25p in two month's time and is currently priced at £5. A six-month forward contract is available on the share for £4.85. The risk-free force of interest is 3% per annum.

(i) Calculate the no-arbitrage price for the forward contract.

(1 mark)

(ii) Show that an investor can make a risk-free profit.

(4 marks)

(Total 6 marks)

PLEASE TURN OVER

Q6 Three bonds, paying annual coupons in arrears of 6%, are redeemable at £105 per £100 nominal. They reach their redemption date in exactly one, two and three years' time respectively. The price of the 1-year bond is £105 per £100 nominal, that of the 2-year bond £104 per £100 nominal and the 3-year bond is priced at £103 per £100 nominal

(a) Calculate all possible spot rates implied by the information above, expressing each of them as an effective annual rate of interest.

(3 marks)

(b) Calculate all possible forward rates implied by the information above, expressing each of them as an effective annual rate of interest.

(4 marks)

(Total 7 marks)

END OF PAPER